

Flow B2 - Controlled Wazuh + AI Security Policy Deployment

A GitHub-approved deployment gate for safely moving validated Wazuh rules/decoders and MCP/RAG/agentic policy bundles into the lab environment with backup, staging, restart, postdeploy proof, rollback readiness, Slack, GitHub, and DataTables.

Document Type: Workflow project document | Flow B2
Prototype Focus: Controlled deployment | Wazuh manager | AI policy bundles | GitHub approval gate
Design Goal: Demonstrate safe deployment of validated AI security content after CI pass, approval, and explicit deployment signal.
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Why this document matters

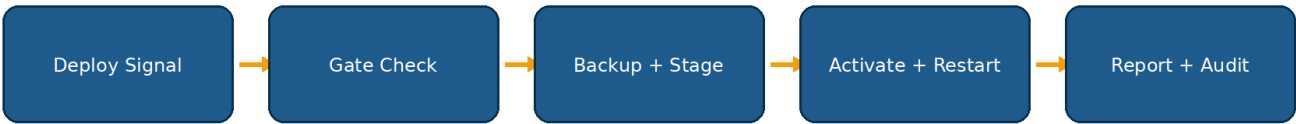
- The write-up is designed as both a portfolio-grade project document and an implementation-facing prototype note.
- It explains the problem, the workflow logic, the SOC value, and the validation evidence captured from real lab executions.
- The project extends the previous MVP V1 pattern into MCP, RAG/memory, and agentic AI security operations.

Key outcome	Why it matters
Blocked unsafe PR	Flow B2 refused deployment for PR #20 because A2 failed and approval requirements were not met.
Successful deployment	Flow B2 deployed PR #19 after A2 pass labels, ready-to-deploy, approved label, and /deploy-lab signal.
Deployment audit	DataTables recorded deployment status and policy bundle hashes.

Architecture view

Flow B2 - Controlled Wazuh + Policy Deployment

Approval-gated deployment with backup, predeploy checks, postdeploy proof, and rollback readiness



MVP V2 prototype: production-inspired design, validated in lab, prepared for portfolio and further hardening.

Architecture view - Flow B2 converts approved GitHub deployment signals into a controlled Wazuh and policy-bundle release lifecycle.

Project Summary

Flow B2 is the release-control layer of MVP V2. It checks PR labels, approval state, and deployment signal before running a controlled deployment sequence. The workflow handles blocked and successful deployments with GitHub comments, Slack notifications, and DataTable evidence.

Problem Statement

Validated content still needs safe release control. Without a deployment gate, a rule or policy bundle could be copied into Wazuh without backup, staging, XML validation, smoke testing, or rollback preparation. That creates avoidable outage and incident-response risk.

Solution Overview

Flow B2 extends the MVP V1 deployment workflow by deploying not only Wazuh rules/decoders but also AI security policy bundles for MCP, RAG/memory, and agentic behavior. It keeps the same safe-release principle but protects the full GenAI runtime stack.

Tools and Technologies Used

Tool	Role in the prototype
GitHub	PR label gate, approval signal, /deploy-lab command, and deployment report comment.
n8n	Gate orchestration, branch handling, local runner execution, Slack, and DataTables.
Wazuh Manager	Receives validated rules/decoders after backup, staging, XML check, and restart.
Python deploy runner	Performs isolated checkout, backup, staging, activation, postdeploy validation, and JSON result output.
Slack	SOC-facing blocked/success deployment status.
DataTables	Deployment run and policy-bundle deployment evidence.

How the System Works

Stage	What happens
Signal	A PR comment /deploy-lab, approved state, or deploy label triggers the workflow.
Gate	Required labels and approval are checked before any Wazuh change happens.
Blocked path	Unsafe PRs produce GitHub/Slack block reports and no Wazuh changes.
Success path	Approved PRs run backup, checkout, stage, XML check, replay smoke, activation, restart, policy bundle activation, and audit.
Audit	Deployment and policy-bundle rows are written for traceability.

SOC Value and Expected Impact

- Reduces MTTR during bad release recovery because backup and rollback paths are part of the workflow.
- Improves change-control consistency by preventing deploys unless A2 pass + approval + explicit signal exist.
- Reduces deployment anxiety by producing clear GitHub/Slack reports and DataTable evidence.
- Supports SOC change governance by tying deployment to PR number, commit SHA, labels, and policy bundle hash.

Implementation and Validation Evidence

The B2 evidence includes a blocked deployment gate test and a successful deployment test. The blocked path proves safety; the success path proves controlled release and auditability.



Figure B2-01 - Full Flow B2 production workflow with GitHub deploy signal, gate check, deploy runner, reports, and DataTables.

PUT `HTTP_GitHub_Post_B2_Blocked_Comment`

Parameters

Method: POST

URL: `https://api.github.com/repos/{{ $json.deploy_context.repo }}/issues/{{ $json.deploy_context.pr_number }}/comments`

Authentication: GitHub API

Credential Type: GitHub API

GitHub API: GitHub account 2

Send Query Parameters: ☐ Send Headers: ☒ Specify Headers: Using Fields Below

Headers:

Accept	Name
Accept	

OUTPUT

Execution will continue even if the node fails

1 Item

```
{
  "organizations_url": "https://api.github.com/users/abdul4rehman215/orgs",
  "repos_url": "https://api.github.com/users/abdul4rehman215/repos",
  "events_url": "https://api.github.com/users/abdul4rehman215/events",
  "received_events_url": "https://api.github.com/users/abdul4rehman215/received_events",
  "type": "User",
  "user_view_type": "public",
  "site_admin": false
},
{
  "created_at": "2026-05-09T13:36:48Z",
  "updated_at": "2026-05-09T13:36:48Z",
  "author_association": "OWNER",
  "body": "## Flow B2 Controlled Mazuh + AI Security Policy Deployment\n\nDecision: ** BLOCKED\n\nDeployment status: ** blocked\n\nCommit: ** 178c739efdb9e4f8ff6f1d9bdcac6d4d0077d8\n\nActor: ** abdul4rehman215\n\nSignal: ** manual_deploy_comment\n\nGate: ** FAIL | approved=false\n\nBlock reason:\n\nBlocked missing labels: ai-security-ci-pass,agentic-policy-pass,ready-to-deploy\n\nRequired labels: ai-security-ci-pass,detection-ci-pass,mcp-policy-pass,rag-policy-pass,agentic-policy-pass,ready-to-deploy\n\nObserved labels: ai-security-ci-fail,mcp-policy-pass,rag-policy-pass,agentic-policy-fail,needs-review,detection-ci-pass\n\nApproval required: true\n\nValid deployment signal required: true\n\nSafety result:\n\nNo Mazuh files were staged, activated, or restarted. Deployment stopped before touching the Mazuh manager.",
  "reactions": {
    "url": "https://api.github.com/repos/abdul4rehman215/wazuh-genai-ci/issues/comments/4412647278/reactions",
    "total_count": 0,
    "+1": 0,
    "-1": 0
  }
}
```

Figure B2-02 - B2 deploy context and gate logic node output for a blocked deployment request.

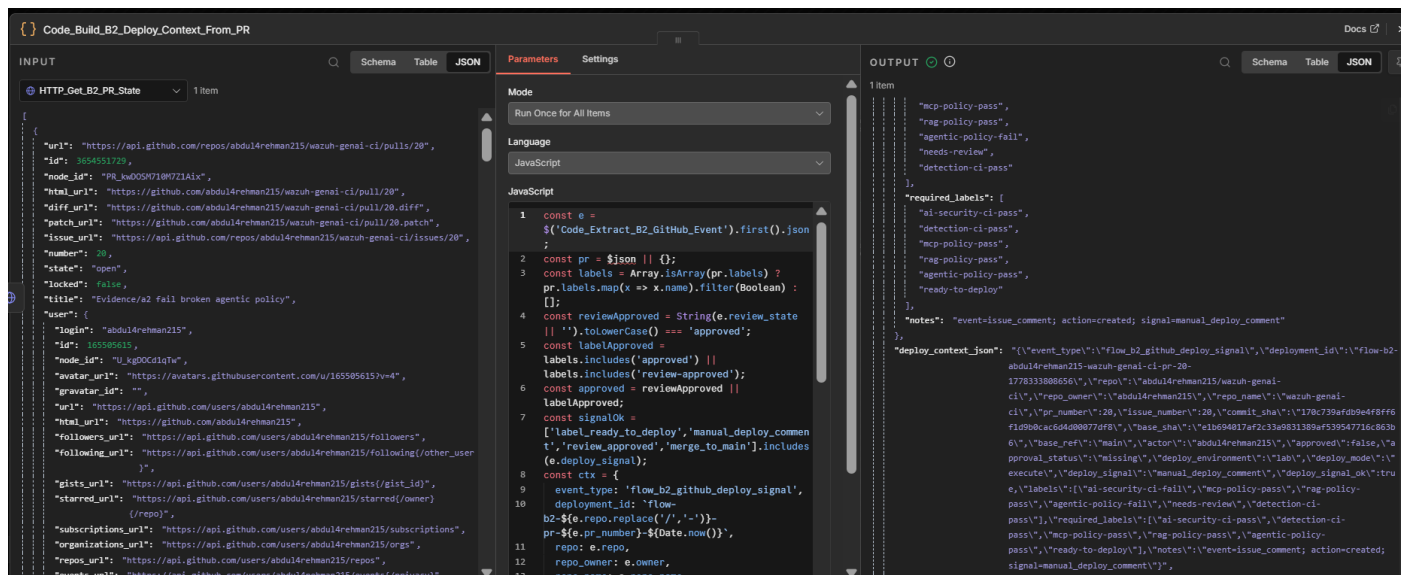


Figure B2-03 - B2 gate state and PR label evidence showing missing requirements for the blocked path.

Evidence/a2 fail broken agentic policy #20
 abdul4rehman215 wants to merge 2 commits into `main` from `evidence/a2-fail-broken-agentic-policy`

`/deploy-lab`

abdul4rehman215 commented now

Flow B2 Controlled Wazuh + AI Security Policy Deployment

Decision: BLOCKED
Deployment status: blocked
Repo/PR: abdul4rehman215/wazuh-genai-ci #20
Commit: 170c739
Actor: abdul4rehman215
Signal: manual_deploy_comment
Gate: FAIL | approved=false

Block reason

blocked missing=ai-security-ci-pass,agentic-policy-pass,ready-to-deploy approval=false

Required gate

- Required labels: ai-security-ci-pass,detection-ci-pass,mcp-policy-pass,rag-policy-pass,agentic-policy-pass,ready-to-deploy
- Observed labels: ai-security-ci-fail,mcp-policy-pass,rag-policy-pass,agentic-policy-fail,needs-review,detection-ci-pass
- Approval required: true
- Valid deploy signal required: true

Safety result

No Wazuh files were staged, activated, or restarted. Deployment stopped before touching the Wazuh manager.

Figure B2-04 - GitHub blocked deployment comment showing missing labels and safety stop before Wazuh activation.

Wazuh-SOC-Alerts APP 19:02
Flow B2 Controlled Deployment BLOCKED
 Repo/PR: abdul4rehman215/wazuh-genai-ci #20
 Commit: 170c739afb9e4f8ff6f1d9b0cac6d4d00077df8
 Signal: manual_deploy_comment
 Reason: blocked missing=ai-security-ci-pass,agentic-policy-pass,ready-to-deploy approval=false
 Observed labels: ai-security-ci-fail,mcp-policy-pass,rag-policy-pass,agentic-policy-fail,needs-review,detection-ci-pass
 Required labels: ai-security-ci-pass,detection-ci-pass,mcp-policy-pass,rag-policy-pass,agentic-policy-pass,ready-to-deploy
 Safety: no Wazuh changes applied
 Automated with this [n8n workflow](#)

Figure B2-05 - Slack blocked-deployment notification with observed labels, required labels, and safety result.

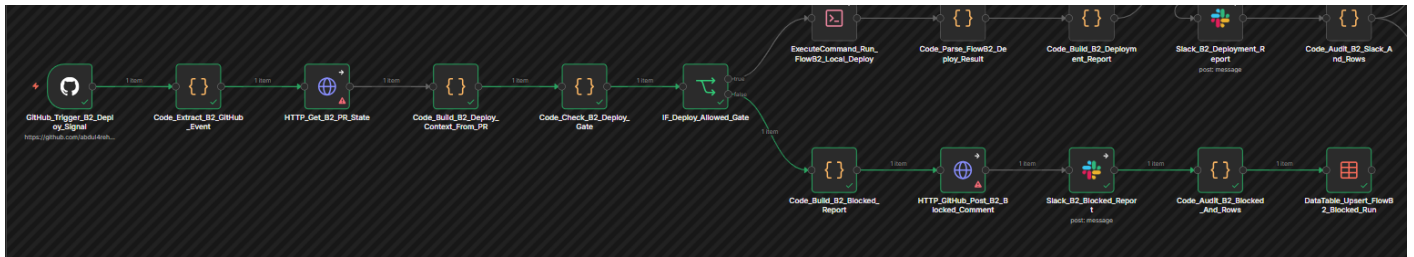


Figure B2-06 - n8n blocked execution path showing gate branch and no deployment runner activation.

Code_Parse_FlowB2_Deploy_Result

Mode: Run Once for All Items

Language: JavaScript

```

1 const ctx =
2   {{(Code_Check_B2_Deploy_Gate).first().json;
3   $json.data || ''}};
4 const rawErr = String($json.stderr ||
5   $json.error || '');
6 let result = null;
7 try {
8   result = JSON.parse(rawErr || '{}');
9 } catch (e1) {
10   const start = rawErr.indexOf('{');
11   const end = rawErr.lastIndexOf('}');
12   result = start >= 0 && end > start ?
13     JSON.parse(rawErr.slice(start, end + 1)) :
14     null;
15 } catch (e2) {
16   result = null;
17 }
18 if ((result || typeof result !== 'object')) {
19   result = {
20     "deployment_id": "flow-b2-abdul4rehman215-wazuh-genai-ci-pr-19-1778334055339",
21     "repo": "abdul4rehman215/wazuh-genai-ci",
22     "repo_owner": "abdul4rehman215",
23     "repo_name": "wazuh-genai-ci",
24     "pr_number": 19,
25     "issue_number": 19,
26     "commit_sha": "345e72b7f4ab784d3eaf6b27a653fc016180ff8",
27     "base_sha": "c1b694017ef2c33a9831389af539547716c663b6",
28     "base_ref": "main",
29     "actor": "abdul4rehman215",
30     "approved": true,
31     "approval_status": "approved",
32     "deploy_environment": "lab",
33     "deploy_mode": "execute",
34     "deploy_signal": "manual_deploy_comment",
35     "deploy_signal_ok": true,
36     "labels": [
37       "approved",
38       "mcp-policy-pass",
39       "rag-policy-pass",
40       "needs-review",
41       "agentic-policy-pass",
42       "ai-security-ci-pass",
43       "detection-ci-pass",
44       "ready-to-deploy"
45     ],
46     "required_labels": [
47       "ai-security-ci-pass",
48       "detection-ci-pass",
49       "mcp-policy-pass",
50       "rag-policy-pass",
51       "agentic-policy-pass",
52       "ready-to-deploy"
53     ]
54   };
55 }
56 
```

Figure B2-07 - B2 success-path node output showing approved PR context and deployment row construction.

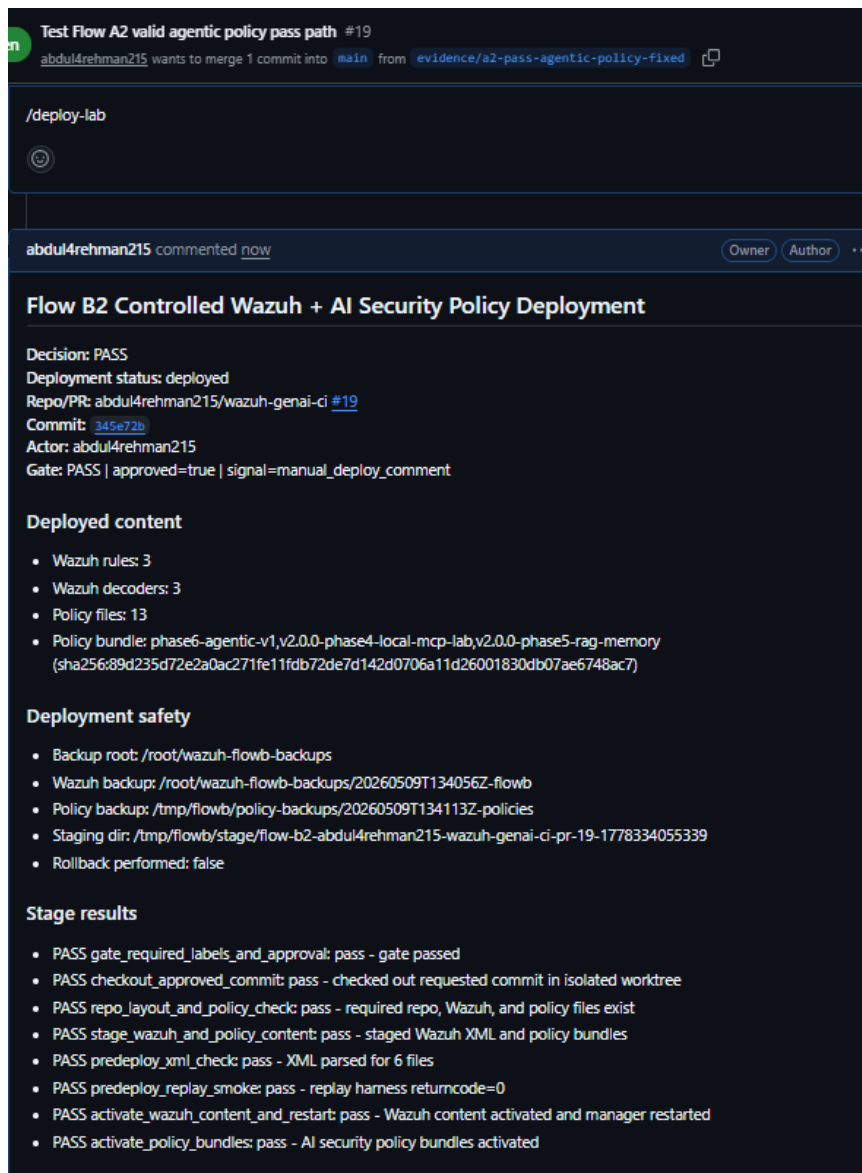


Figure B2-08 - GitHub success deployment report for PR #19 with 8/8 deployment stages passed.

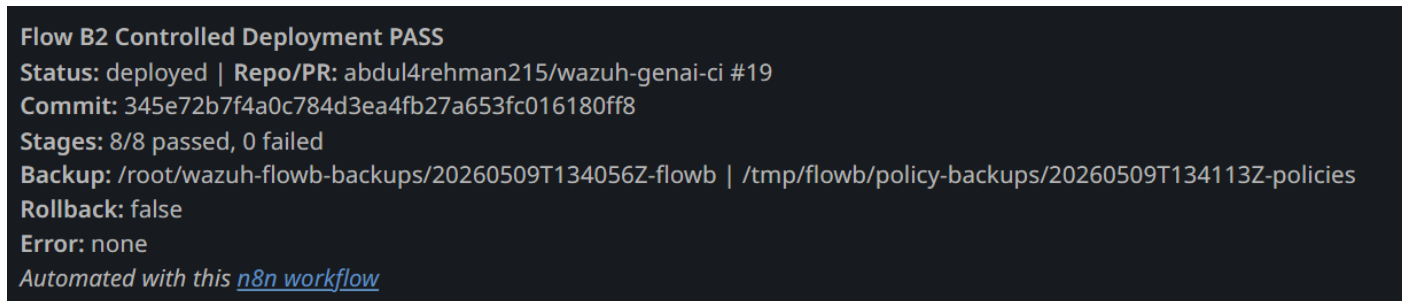


Figure B2-09 - Slack deployment PASS notification showing deployed status, backup path, and rollback state.

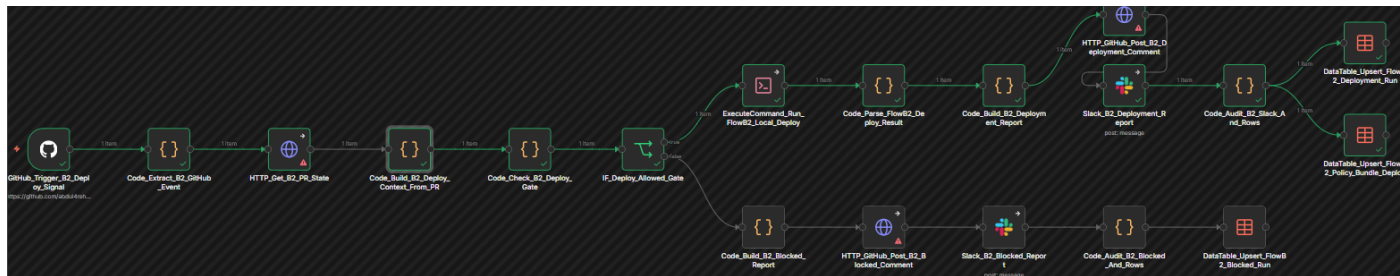


Figure B2-10 - n8n success execution path through deploy runner, GitHub comment, Slack, and audit nodes.

deployment_run_id	timestamp	T deployment_mode	T deployment_status	T repo	# pr_number	# commit_sha	T actor	approved	gate_passed	T required_labels_csv	T observed_labels_csv
abdul4rehman215-wazuh-gen	2026-05-09T19:02	execute	blocked	abdul4rehman215/wazuh-gen	20	Null	abdul4rehman215			ai-security-ci-pass,detection-ci-pass,n	ai-security-ci-fail,mcp-policy
abdul4rehman215-wazuh-gen	2026-05-09T19:11	execute	deployed	abdul4rehman215/wazuh-gen	19	Null	abdul4rehman215			ai-security-ci-pass,detection-ci-pass,n	approved,mcp-policy

Figure B2-11 - flow_b2_deployment_runs DataTable showing blocked and deployed deployment runs.

T predeploy_status	T postdeploy_status	rollback_performed	T rollback_reason	slack_notified	github_comment_posted	T error	T stage_result
blocked_before_predeploy	not_started		Empty			blocked missing=ai-security-ci-pass,n	{{"stage_name": "predeploy"}}
pass	pass		Empty			Empty	{{"stage": "gate"}}

Figure B2-12 - flow_b2_deployment_runs extended columns with predeploy/postdeploy status, Slack, GitHub, and error values.

T policy_deployment_id	timestamp	T deployment_run_id	T repo	T actor	T policy_bundle_version	T policy_bundle_hash
flow-b2-abdul4rehman215-wazuh-gen	2026-05-09T19:11:13.000+05:30	flow-b2-abdul4rehman215-wazuh-gen	abdul4rehman215/wazuh-gen-ai-ci	abdul4rehman215	phase6-agentic-v1/v2.0.0-phase4-local	sha256:89d235d72e2a0ac271fe11fdb7

Figure B2-13 - flow_v2_policy_bundle_deployments DataTable showing policy bundle version and hash.

# mcp_policy_files_count	# rag_memory_policy_files_count	# agentic_policy_files_count	T policy_target_dir	T policy_backup_path	T policy_staging_path	T deployment_status
4	4	5	/tmp/flowb/active-policies	/tmp/flowb/policy-backups/202605091	/tmp/flowb/stage/flow-b2-abdul4rehman215	deployed

Figure B2-14 - Policy bundle deployment counts for MCP, RAG/memory, and agentic policy files.

T active_mcp_bundle_hash	T active_rag_memory_bundle_hash	T active_agentic_bundle_hash	T postdeploy_validation_status	T error	T notes
sha256:c3348bab344a682eee6456a51	sha256:689405655db379593cca11fc4e	sha256:990ee4d66767b6c35add22721	pass	Empty	Flow B2 deployed Wazuh XML plus MCP

Figure B2-15 - Policy deployment postdeploy validation status and active bundle hashes.

Limitations and Production Hardening Roadmap

- The prototype uses lab SSH and local runner patterns; production should move to a GitHub App, signed deploy identities, vault-backed secrets, and stronger RBAC.
- Rollback logic is implemented conceptually and validated in lab, but production use should add Wazuh health checks, automatic rollback triggers, and canary deployment.
- Additional guardrails should include approval quorum, environment locks, immutable audit logs, and change-window controls.

Skills Demonstrated

Skill area	Evidence in this build
Release engineering	Built a gated deployment lifecycle with backup, staging, validation, activation, restart, and postdeploy status.
SOAR automation	Implemented GitHub-triggered n8n orchestration with Slack and audit outputs.
AI security policy deployment	Deployed MCP, RAG/memory, and agentic policy bundles alongside Wazuh content.
Troubleshooting	Resolved command handoff, DataTable schema, local/remote path, Git checkout, and SSH activation issues.

Conclusion

Flow B2 proves that MVP V2 has a controlled release path, not only detections. It prevents unsafe deployment and provides visible evidence for safe releases.

Together, A2 and B2 form the detection-as-code and policy-as-code engineering backbone of the project.

Document prepared for portfolio showcase by [abdul4rehman215](#)